An explicit superpolynomial upper bound for the majority-consensus threshold

A proof note based on Kanoria–Montanari (2011)

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Abstract

For the synchronous majority dynamics and fair tie-breaking convention of Kanoria–Montanari, this note derives the upper bound

$$\theta_*(k) \le \exp\left\{-\frac{\log k}{\log 2}\log\log k + O(\log k)\right\}.$$

The argument makes the time horizon in the dynamic cavity calculation quantitative. Its two main ingredients are bounds on the covariance and response of the cavity process, and a one-coordinate local central limit estimate that does not involve the least probable trajectory. This is an upper bound, not an asymptotic equivalence; no matching large-degree lower bound is asserted.

1 Statement and relationship to the original paper

All logarithms are natural. The degree is denoted by k; the initial spin mean is θ, so that $\mathbb{P}(\sigma_v(0) = +1) = (1+\theta)/2$. The update is synchronous majority, with a fresh fair output coin at a tie. This has the same law as the tie-breaking convention in [1]. We retain the paper's definitions of $C(t,s)$, $R(t,s)$ and the unbiased cavity process τ:

$$\tau(t+1) = \operatorname{sign}\left(\eta(t) + \sum_{s<t} R(t,s)\tau(s)\right), \qquad \mathbb{E}\eta(t)\eta(s) = C(t,s),$$

where $\tau(0)$ is a fair sign independent of the Gaussian process η. The consistency conditions are

$$C(t,s) = \mathbb{E}\tau(t)\tau(s), \qquad R(t,s) = \partial_{h(s)}\mathbb{E}\tau_h(t)\Big|_{h=0}.$$

Put $C_T = (C(t,s))_{0\le t,s\le T}$, $R_T = (R(t,s))_{0\le t,s\le T}$, with zero entries for $s \ge t$, and

$$r_j = R(j, j-1), \qquad a_0 = 1, \qquad a_t = \prod_{j=1}^{t} r_j.$$

The existence, strict positive definiteness, and finite-time cavity identities proved in [1] are used below. The proof does not substitute a growing time into an unquantified fixed-time convergence theorem.

Theorem 1.1. *There are absolute constants A, B, k_0 such that the following holds. Let $T \ge 0$ and $k \ge k_0$ satisfy*

$$\log k \ge A 2^{T/2}. \tag{1}$$

For

$$\theta = \frac{8}{a_T k^{(T+1)/2}}, \tag{2}$$

one has $0<\theta<1$ *and*

$$\mathbb{P}_\theta\{\sigma_v(T+2)=-1\}\le e^{-k/8}.$$

Consequently,

$$\theta_*(k)\le\frac{8}{a_T k^{(T+1)/2}}\le k^{-(T+1)/2}\exp\{B2^{T/2}\}. \tag{3}$$

The conclusion holds for odd and even degrees.

Theorem 1.2. *As* $k\to\infty$*,*

$$\theta_*(k)\le\exp\left\{-\frac{\log k}{\log 2}\log\log k+O(\log k)\right\}. \tag{4}$$

In particular, for every $c<1/\log 2$ *and every sufficiently large* k*,*

$$\theta_*(k)\le e^{-c\log k\log\log k}.$$

The original Theorem 2.3 in [1] gives $\theta_*(k)=o(k^{-M})$ for every fixed M. The improvement in (4) is an explicit choice of a diverging exponent. The constant $1/\log 2$ is a constant produced by this argument; it is not claimed to be the true asymptotic constant of the threshold.

2 Structural facts about the cavity process

Lemma 2.1. *Every finite cavity trajectory is associated and invariant under simultaneous spin reversal. Moreover,*

$$C(t,s)\ge 0,\qquad R(t,s)\ge 0,$$

and

$$C(t,s)=0\quad(t-s\ \textit{odd}),\qquad R(t,s)=0\quad(t-s\ \textit{even}). \tag{5}$$

The even and odd spin subsequences are independent. For every nonempty finite set J *of time indices,*

$$\mathbb{P}\{\tau(s)=+1\ \textit{for all}\ s\in J\}\ge 2^{-|J|},\qquad C_J\succeq 2^{1-|J|}\mathbf{1}_J\mathbf{1}_J^\top. \tag{6}$$

Proof. Induct on time. If the previously constructed responses are nonnegative, the effective-process spins are nondecreasing functions of the past Gaussian fields, the initial sign, and the added fields h. Thus the newly constructed responses are nonnegative. A Gaussian vector with nonnegative covariances is associated [2]; adjoining an independent initial sign preserves association. Applying nondecreasing maps therefore shows that the enlarged spin trajectory is associated, and hence that its covariances are nonnegative. Simultaneous spin reversal preserves its law and gives zero means.

The parity identities follow from the recursion. The odd spins depend only on the even-indexed Gaussian fields. The even spins depend only on $\tau(0)$ and the odd-indexed Gaussian fields. The two Gaussian blocks are independent, giving the asserted independence and completing the parity induction. This is the bipartite decomposition noted in [1].

Association gives the first inequality in (6). The two constant sign vectors have equal probabilities. Their contribution to $\mathbb{E}\tau_J\tau_J^\top$ gives the second inequality. □

Lemma 2.2 (Gaussian projection inequality)**.** *Let* $\boldsymbol{c}_T=(C(t,0))_{0\le t\le T}$*. Then*

$$C_T\succeq \boldsymbol{c}_T\boldsymbol{c}_T^\top+R_TC_TR_T^\top. \tag{7}$$

In particular,

$$\mathbb{E}\left(\sum_{s<t}R(t,s)\tau(s)\right)^2\le 1,\qquad \sum_{s<t}R(t,s)^2\le 1,\qquad \sum_{s<t}R(t,s)\le\sqrt{t}. \tag{8}$$

Proof. Shifting the Gaussian vector by h is exactly the same as adding h in the effective-process recursion. Gaussian differentiation therefore gives

$$\mathbb{E}[\tau^T(\eta^T)^\top] = R_T C_T.$$

The orthogonal projection of τ^T onto the linear span of $\tau(0), \eta(0), \ldots, \eta(T)$ is $\boldsymbol{c}_T\tau(0) + R_T\eta^T$. The covariance matrix of its residual is positive semidefinite, proving (7). Its diagonal gives the first inequality in (8). Since R and C are entrywise nonnegative and $C(s,s) = 1$, it also gives $\sum_s R(t,s)^2 \le 1$. Cauchy–Schwarz proves the final assertion. □

3 A Gaussian boundary-mass estimate

The following lemma is the device that avoids a minimum-trajectory probability. A rare, arbitrary sign pattern is not used.

Lemma 3.1. *Let $X \in \{-1,+1\}^m$, let $C = \mathbb{E}XX^\top$ be positive definite, and suppose*

$$p = \mathbb{P}\{X_1 = \cdots = X_m\} > 0.$$

For $G \sim \mathcal{N}(0, C)$ and $m \ge 2$,

$$\mathbb{P}\{G_1, \ldots, G_{m-1} \ge 0 \mid G_m = 0\} \ge \frac{e^{-1}p^{(m-2)/2}}{2^{3(m-1)/2}\Gamma((m+1)/2)}. \tag{9}$$

For $m = 1$ the conditional orthant probability is understood as 1. Furthermore, writing $\phi_{\mu,C}$ for Gaussian density and $L = \mu^\top C^{-1}\mu$, one has

$$\int_{\mathbb{R}_+^{m-1}} \phi_{\mu,C}(z_1, \ldots, z_{m-1}, 0)\, dz \ge (2\pi)^{-1/2}e^{-L}\left[\mathbb{P}\{G_{<m} \ge 0 \mid G_m = 0\}\right]^2. \tag{10}$$

Proof. Set $D_i = (1 - X_iX_m)/2$ for $i < m$, $q = \mathbb{E}D$, and $M = \mathbb{E}DD^\top$. A direct calculation gives

$$C_{<m,<m} - C_{<m,m}C_{m,<m} = 4(M - qq^\top).$$

This matrix is positive definite. Consider jointly centered Gaussian (W, Z) with covariance

$$\begin{pmatrix} M & q \\ q^\top & 1 \end{pmatrix}.$$

This covariance matrix is the Gram matrix of the random vector $(D, 1)$. Let $v = 1 - q^\top M^{-1}q$. For every deterministic vector a,

$$\mathbb{E}(1 - a^\top D)^2 \ge \mathbb{P}(D = 0) = p,$$

so that $v \ge p$. Conditional on W, Z has mean $h(W) = q^\top M^{-1}W$ and variance v. The variance of $h(W)$ is $1 - v \le 1$.

Write $W = M^{1/2}Z_0$, where Z_0 is a standard Gaussian vector in dimension $d = m-1$. Then $h(W) = a^\top Z_0$ for a vector with $\|a\|^2 = 1 - v \le 1$. The entries of M are nonnegative, so Gaussian association gives $\mathbb{P}(W \ge 0) \ge 2^{-d}$. The event $W \ge 0$ is a cone in Z_0, and hence is independent of the Gaussian radius $\|Z_0\|$. On $\{\|Z_0\| \le \sqrt{v}\}$, the density ratio for conditioning $Z = 0$ satisfies

$$\frac{f_{Z|W}(0 \mid W)}{f_Z(0)} = v^{-1/2}\exp\{-h(W)^2/(2v)\} \ge v^{-1/2}e^{-1/2}.$$

Since $0 < v \le 1$, integration of the standard Gaussian density over the ball of radius $\sqrt{v}$ gives

$$\mathbb{P}(\|Z_0\| \le \sqrt{v}) \ge \frac{e^{-1/2} v^{d/2}}{2^{d/2}\Gamma(d/2+1)}.$$

Combining these facts and using $v \ge p$ yields

$$\mathbb{P}(W \ge 0 \mid Z = 0) \ge \frac{e^{-1} v^{(d-1)/2}}{2^{3d/2}\Gamma(d/2+1)} \ge \frac{e^{-1} p^{(m-2)/2}}{2^{3(m-1)/2}\Gamma((m+1)/2)}.$$

The conditional covariance of W is $M - qq^\top$; multiplication by 2 identifies its orthant probability with the one in (9).

For the second assertion, put $\Gamma = C_{<m,<m} - C_{<m,m} C_{m,<m}$ and $\nu = \mu_{<m} - C_{<m,m}\mu_m$. The conditional mean and covariance on the last-coordinate-zero slice are ν, Γ, and

$$L = \mu_m^2 + \nu^\top \Gamma^{-1} \nu.$$

For any measurable A, Cauchy–Schwarz applied to the Gaussian likelihood ratio gives

$$\mathcal{N}(\nu, \Gamma)(A) \ge \mathcal{N}(0, \Gamma)(A)^2 e^{-\nu^\top \Gamma^{-1} \nu}.$$

Multiplying by the marginal density $(2\pi)^{-1/2} e^{-\mu_m^2/2}$ proves (10). $\square$

4 Quantitative bounds on response and covariance

Proposition 4.1. *There is an absolute constant B such that, for every $t \ge 1$,*

$$0 < r_t \le 1, \qquad r_t \ge \exp\{-B 2^{t/2}\}, \tag{11}$$

and, for every $T \ge 0$,

$$\lambda_{\min}(C_T) \ge \exp\{-B 2^{T/2}\}, \qquad a_T \ge \exp\{-B 2^{T/2}\}. \tag{12}$$

Proof. First consider $r_{t+1} = R(t+1, t)$. Let

$$I = \{s \le t : s \equiv t \pmod 2\}, \qquad J = \{s < t : s \not\equiv t \pmod 2\}.$$

Set $n = |I| = \lfloor t/2 \rfloor + 1$ and $m = |J| = \lceil t/2 \rceil$. To lower-bound the last-step response, retain only the trajectory in which all the past spins in the active parity class are $+1$. On that trajectory the Gaussian field vector on I has deterministic shift

$$\mu = R_{I,J} \mathbf{1}_J.$$

If J is empty, $\mu = 0$. Otherwise, (7) and (6) imply

$$C_I \succeq R_{I,J} C_J R_{I,J}^\top \succeq 2^{1-m} \mu\mu^\top, \qquad \mu^\top C_I^{-1} \mu \le 2^{m-1}.$$

The spin vector τ_I satisfies $\mathbb{P}(\tau_I \text{ is constant}) \ge 2^{1-n}$. Lemma 3.1 gives a conditional Gaussian orthant probability of at least $\exp\{-B(n+1)^2\}$. Its shifted-face bound and $\mu^\top C_I^{-1} \mu \le 2^{m-1}$ therefore lower-bound the Gaussian face mass for this trajectory by

$$\exp\{-B(2^m + (n+1)^2)\} \ge \exp\{-B' 2^{t/2}\}.$$

Here the empty-J case has zero shift cost and is included by enlarging the absolute constants. This face mass is a lower bound on r_{t+1}: when the active class contains $\tau(0)$ its probability

1/2 is canceled by the factor 2 in differentiating the expectation of a sign; otherwise the factor is 2 and may simply be discarded. This proves the lower bound in (11), after changing the absolute constant. The upper bound follows from (8).

For the covariance bound, iterate (7), using the nilpotence of R_T:

$$C_T \succeq \sum_{j=0}^{T} R_T^j \boldsymbol{c}_T \boldsymbol{c}_T^\top (R_T^\top)^j = V_T V_T^\top, \qquad V_T = (R_T^j \boldsymbol{c}_T)_{j=0}^T.$$

The displayed vectors are the columns of V_T. This matrix is lower triangular and its jth diagonal entry, with indexing starting at 0, is a_j. Consequently,

$$\det C_T \geq \prod_{j=0}^{T} a_j^2 = \prod_{j=1}^{T} r_j^{2(T+1-j)}.$$

Since $\lambda_{\max}(C_T) \leq T+1$,

$$-\log \lambda_{\min}(C_T) \leq T\log(T+1) + 2\sum_{j=1}^{T}(T+1-j)(-\log r_j).$$

The geometric-sum estimates

$$\sum_{j=1}^{T} 2^{j/2} \leq B2^{T/2}, \qquad \sum_{j=1}^{T}(T+1-j)2^{j/2} \leq B2^{T/2}$$

prove both assertions in (12). □

Remark 4.2. The factor $2^{T/2}$, rather than 2^T, is important. The boundary calculation involves one parity block. Discarding this decomposition would lose a factor 2 in the leading coefficient of (4).

5 A one-coordinate local central limit estimate

The original local CLT assumes a lower bound on every trajectory atom. Here only one coordinate is pinned, allowing that assumption to be removed.

Lemma 5.1. *Let $X_1, \ldots, X_N$ be i.i.d. in $\{-1,+1\}^d$, with mean m and covariance Σ, where*

$$\left\|\sqrt{N}m\right\|_\infty \leq M, \qquad \Sigma \succeq \lambda I, \qquad 0 < \lambda \leq 1.$$

Let $S = \sum_i X_i$, $\mu = \sqrt{N}m$ and $G \sim \mathcal{N}(\mu, \Sigma)$. There is a fixed numerical exponent c_0 such that, with

$$H = \left\lceil \frac{C(d+1)(M+1)}{\lambda} \right\rceil^{c_0},$$

the following estimates hold whenever $N \geq H^2$. Ordinary orthant probabilities, with integer thresholds of absolute value at most 5, have Gaussian approximation error at most $H/\sqrt{N}$, after normalizing S by $\sqrt{N}$. For any coordinate j, any admissible parity value b with $|b| \leq 5$, and signs $\varepsilon_i \in \{-1,+1\}$,

$$\left| \frac{\sqrt{N}}{2}\mathbb{P}\{S_j = b,\ \varepsilon_i S_i \geq b_i\ (i \neq j)\} - \int_{\substack{z_j=0 \\ \varepsilon_i z_i \geq 0\ (i\neq j)}} \phi_{\mu,\Sigma}(z)\, dz_{-j} \right| \leq \frac{H}{\sqrt{N}}, \tag{13}$$

where $|b_i| \leq 5$. The same statement holds with strict inequalities and with averages of strict and weak inequalities. Gaussian orthant probabilities and these Gaussian face integrals are Lipschitz in their mean and covariance on the indicated parameter range, with a constant bounded by a fixed polynomial in $d+1, M+1, \lambda^{-1}$.

Proof. The ordinary approximation follows from the convex-set multivariate Berry–Esseen bound [3]. After whitening, each centered summand has norm at most $2\sqrt{d/\lambda}$, giving an error at most $Cd^{7/4}\lambda^{-3/2}/\sqrt{N}$.

For (13), condition on $S_j = b$. There are exactly $N_+ = (N + b)/2$ summands with jth coordinate $+1$ and $N_- = (N - b)/2$ with jth coordinate -1. Conditional on these counts, the sum of the remaining coordinates has the distribution of a sum of N_+ independent vectors with law $X_{-j} \mid X_j = +1$ and N_- independent vectors with law $X_{-j} \mid X_j = -1$. It does not matter which summand indices have the two signs.

Write $p_\pm = \mathbb{P}(X_j = \pm 1)$ and let $\Sigma_\pm$ be these two conditional covariances. The weighted average

$$p_+\Sigma_+ + p_-\Sigma_- = \Sigma_{-j,-j} - \frac{\Sigma_{-j,j}\Sigma_{j,-j}}{\Sigma_{jj}} =: \Gamma$$

is a Schur complement and satisfies $\Gamma \succeq \lambda I$. Changing the weights from $p_\pm$ to $N_\pm/N$ changes this matrix in operator norm by at most $Cd(M+1)/\sqrt{N}$. Thus the conditional sum covariance is at least $N\lambda I/2$ under the stated size assumption. Apply the independent, not necessarily identically distributed version of the same Berry–Esseen bound.

Since X_j is binary, its conditional regression is exactly affine. The conditional mean of the normalized sum is therefore

$$\mu_{-j} + \frac{\Sigma_{-j,j}}{\Sigma_{jj}}(b/\sqrt{N} - \mu_j),$$

which is the corresponding Gaussian conditional mean. The binomial local estimate, obtained directly from Stirling's formula, is

$$\mathbb{P}(S_j = b) = \frac{2}{\sqrt{N}}\phi_{\mu_j,\Sigma_{jj}}(b/\sqrt{N}) + O(H/N).$$

Multiplying this estimate by the conditional orthant approximation proves (13). Moving the bounded thresholds to zero has the same order of error.

For completeness, covariance and mean comparison can be done on Gaussian densities before integration. A mean derivative contributes a linear Gaussian score and a covariance derivative contributes a quadratic score. Their absolute integrals on an orthant are bounded by their full-space absolute integrals. On a coordinate-zero face, condition on that coordinate first. The conditional mean and covariance are the affine regression and Schur complement above; their inverse norms are bounded by a fixed polynomial in λ^{-1}, and their means by a fixed polynomial in $M+1, \lambda^{-1}$. The first two conditional Gaussian moments consequently give the asserted polynomial Lipschitz bounds. All powers needed in this proof are fixed numbers, so one may enlarge c_0 once to cover all estimates. $\square$

6 Kernel calculus without high-order product expansions

For $\ell \in \{-1, +1\}$ define

$$\kappa_\ell(z) = \mathbf{1}\{\ell z > 0\} + \tfrac{1}{2}\mathbf{1}\{z = 0\}.$$

For $\lambda = (\lambda_0, \ldots, \lambda_d)$ and $u = (u_0, \ldots, u_{d-1})$, put

$$F_{\lambda,u}(z) = \prod_{s=0}^{d-1} \kappa_{\lambda_{s+1}}(z_s + u_s).$$

For a Gaussian mean μ and covariance Σ, write

$$J_s(\lambda; \mu, \Sigma) = \int_{\substack{z_s = 0 \\ \lambda_{r+1} z_r \geq 0\ (r \neq s)}} \phi_{\mu,\Sigma}(z)\, dz_{-s}.$$

In the cavity process at dimension $d = t+1$, use the abbreviation

$$\mu_r(\lambda) = \sum_{v<r} R(r,v)\lambda_v, \qquad J_s^t(\lambda) = J_s(\lambda; \mu(\lambda), C_t).$$

Gaussian differentiation of the path formula gives

$$R(t+1, s) = \frac{1}{2} \sum_{\lambda \in \{-1,+1\}^{t+2}} \lambda_{t+1}\lambda_{s+1} J_s^t(\lambda). \tag{14}$$

In particular J_t^t is independent of λ_{t+1} and $\sum_{\lambda_0,\dots,\lambda_t} J_t^t(\lambda) = r_{t+1}$. Also $0 \le J_s^t(\lambda) \le (2\pi)^{-1/2}$.

Lemma 6.1. *Under the hypotheses of Lemma 5.1, changing a bounded integer field u to v gives*

$$\mathbb{E}[F_{\lambda,v}(S) - F_{\lambda,u}(S)] = \frac{1}{\sqrt{N}} \sum_s (v_s - u_s)\lambda_{s+1} J_s(\lambda; \mu, \Sigma) + O\left(\frac{H}{N} \|v-u\|_1\right).$$

Here all field coordinates have absolute value at most 2. For two stochastically ordered laws $P \preceq Q$ on $\{-1,+1\}^d$, let $P_\alpha = (1-\alpha)P + \alpha Q$, and put $\Delta_s = \mathbb{E}_Q X_s - \mathbb{E}_P X_s \ge 0$. The exact interpolation identity is

$$\begin{aligned} &\mathbb{E}_{Q^{\otimes n}} F\left(\sum_i X_i\right) - \mathbb{E}_{P^{\otimes n}} F\left(\sum_i X_i\right) \\ &\qquad = n \int_0^1 \mathbb{E}\left[F(S_\alpha + Y) - F(S_\alpha + X)\right] d\alpha, \end{aligned} \tag{15}$$

where (X,Y) is any monotone coupling of P, Q, independent of S_α, and S_α is the sum of $n-1$ i.i.d. P_α vectors. Consequently its leading term is

$$\sqrt{n} \sum_s \Delta_s \lambda_{s+1} \int_0^1 J_s(\lambda; \mu_\alpha, \Sigma_\alpha)\, d\alpha,$$

with error bounded by $CH \sum_s \Delta_s$ when the means and covariances throughout the interpolation satisfy the local CLT bounds. Replacing $n/\sqrt{n-1}$ by $\sqrt{n}$ is absorbed in this error.

Proof. Telescope the product defining F over its time coordinates. At a changed coordinate the difference of the two majority kernels is supported on a bounded number of lattice levels. Its signed total weight on the lattice is $\lambda_{s+1}(v_s - u_s)/2$. Each level has the $2/\sqrt{N}$ prefactor in (13). At other coordinates the factors are strict or weak half-line indicators or averages of these, with bounded threshold shifts. Apply (13) to each term. This proves the first assertion; it also covers a change of a child trajectory by taking fields $u + X, u + Y$.

Differentiate $P_\alpha^{\otimes n}$ and use permutation symmetry of the summands to obtain (15). Under the monotone coupling, $\mathbb{E}\|Y - X\|_1 = \sum_s \Delta_s$. Apply the first assertion conditionally on X, Y and integrate. No division by a path probability is made, and no expansion into terms with two or more replaced children is needed. □

7 Uniform finite-degree cavity estimates

Let $P_{k,\theta}^u$ denote the root trajectory law in the rooted tree with $k-1$ children and imposed parent trajectory u, as in [1]. Only the displayed finite prefix of this law is used. Parent coordinates may be $-1, 0, +1$. For a prescribed root trajectory λ, the exact cavity recursion is

$$P_{k,\theta}^u(\lambda) = \frac{1+\theta\lambda_0}{2} \mathbb{E}\left[F_{\lambda,u}\left(\sum_{i=1}^{k-1} X_i\right)\right], \tag{16}$$

where the X_i are independent with law $P^{\lambda}_{k,\theta}$ on the appropriate shorter child trajectories. The notation denotes dynamics under an imposed trajectory, not conditioning on an observed trajectory.

In the estimates below, a constant of the form

$$\exp\{C2^{T/2}\} \tag{17}$$

will be called a controlled constant. Products of a fixed number of such constants, multiplication by $2^{CT}(T+1)^C$, and sums over $t \le T$ remain of this form. Also, if $L_t \le \exp\{C2^{t/2}\}$, then

$$\prod_{t=0}^{T}(2L_t) \le \exp\{C'2^{T/2}\}. \tag{18}$$

This follows by summing a geometric series; it does not cost an extra factor T in the exponent.

Proposition 7.1. *There is an absolute C such that, for all T and sufficiently large k with $\log k \ge C2^{T/2}$, uniformly in u,*

$$\left\| P^u_{k,0}(\sigma^T \in \cdot) - \mathcal{L}(\tau^T) \right\|_{\mathrm{TV}} \le \frac{\exp\{C2^{T/2}\}}{\sqrt{k}}, \tag{19}$$

$$\max_{t \le T} \left| \sqrt{k}\, \mathbb{E}_{P^u_{k,0}} \sigma(t) - \sum_{s<t} R(t,s) u_s \right| \le \frac{\exp\{C2^{T/2}\}}{\sqrt{k}}. \tag{20}$$

Proof. Let e_t be the supremum over u of the sum of the two errors on the left sides, at horizon t. Initially $e_0 = 0$. For prescribed λ, the child means in (16) obey

$$\sqrt{k-1}\, \mathbb{E} X_s = \mu_s(\lambda) + O(e_t + (t+1)/k).$$

Their covariance differs from C_t entrywise by $O(e_t + (t+1)/k)$. This follows from the total-variation bound on raw second moments and the $O(\sqrt{t+1}/\sqrt{k})$ bound on the means. Under a bootstrap $e_t \le \lambda_{\min}(C_t)/(100(t+1))$, the covariance is at least $C_t/2$ in the needed eigenvalue sense and the normalized means are bounded by $2(t+2)$.

Apply the ordinary orthant CLT in (16), followed by Gaussian parameter comparison. Summing over 2^{t+2} root trajectories bounds the new total-variation error by $L_t(e_t + k^{-1/2})$, where

$$L_t \le 2^{C(t+1)} \left[\frac{C(t+2)}{\lambda_{\min}(C_t)} \right]^{c_1} \le \exp\{C'2^{t/2}\}$$

for a fixed exponent c_1.

For the normalized mean, subtract the recursion with $u = 0$ from that with u. The zero-field mean is exactly zero by spin-flip symmetry. The first part of Lemma 6.1, multiplied by the initial factor $1/2$, gives the leading term

$$\frac{1}{2\sqrt{k}} \sum_s u_s \lambda_{s+1} J^t_s(\lambda)$$

for the trajectory-probability difference. Multiply by λ_{t+1}, sum over trajectories, and use (14). The result is

$$k^{-1/2} \sum_s R(t+1, s) u_s.$$

The scaled error is at most $L_t(e_t + k^{-1/2})$. Thus, after increasing L_t,

$$e_{t+1} \le L_t(e_t + k^{-1/2}).$$

Equation (18) proves the claimed estimates. The degree condition, with a sufficiently large absolute constant, makes every obtained e_t smaller than the bootstrap threshold and satisfies the size requirements in the local CLT. This closes the induction. $\square$

Proposition 7.2 (Uniform weak-bias amplification)**.** *Fix T and let θ be as in* (2)*. Under a degree condition of the form* (1)*, uniformly for $t \le T$ and imposed parent trajectories u,*

$$d_t(u) := \mathbb{E}_{P^u_{k,\theta}}\sigma(t) - \mathbb{E}_{P^u_{k,0}}\sigma(t) = \theta a_t k^{t/2}(1+\varepsilon_t(u)), \qquad |\varepsilon_t(u)| \le \frac{\exp\{C2^{T/2}\}}{\sqrt{k}}. \tag{21}$$

Proof. Write $b_t = \theta a_t k^{t/2}$ and $\Lambda_T = \max(1, \max_{j\le T} r_j^{-1})$. By Proposition 4.1, Λ_T is a controlled constant. Increasing the degree constant ensures $\sqrt{k} \ge 2\Lambda_T$. Hence b_t is increasing, and

$$b_T = \frac{8}{\sqrt{k}}, \qquad b_t \le \frac{8\Lambda_T}{k} \ (t<T), \qquad \sum_{s<t} b_s \le \frac{2\Lambda_T}{\sqrt{k}} b_t. \tag{22}$$

The assertion at $t=0$ is exact. Bootstrap that $|d_s(u)/b_s - 1| \le 1/2$ through time $t<T$.

For a prescribed root trajectory λ, apply (15) to the child laws $P = P^\lambda_{k,0}$ and $Q = P^\lambda_{k,\theta}$. They admit a monotone coupling using the same initial uniforms and fair output coins. Put $D_t = \max_{s\le t,u} d_s(u)$. For the interpolated child laws, the normalized-mean difference from $\mu(\lambda)$ is at most

$$C(e_t + \sqrt{k}D_t + k^{-1/2}).$$

The covariance difference from C_t is at most $C(e_t + D_t + (t+1)/k)$ entrywise. Indeed a monotone coupling gives

$$|\mathbb{E}_Q X_s X_r - \mathbb{E}_P X_s X_r| \le d_s(\lambda) + d_r(\lambda),$$

and the mean correction in the covariance has the same bound up to an absolute factor. All interpolated covariances consequently remain uniformly positive definite under the degree condition.

Use the one-face CLT and its Gaussian parameter comparison in the interpolation. Separating off the change of the root's initial law gives

$$P^u_{k,\theta}(\lambda) - P^u_{k,0}(\lambda) = \frac{\sqrt{k}}{2}\sum_{s=0}^{t} d_s(\lambda)\lambda_{s+1}J^t_s(\lambda) + \mathcal{E}_\lambda, \tag{23}$$

$$|\mathcal{E}_\lambda| \le \frac{\theta}{2} + L_t\sqrt{k}\left(\sum_{s=0}^{t} d_s(\lambda)\right)(e_t + \sqrt{k}D_t + k^{-1/2}), \tag{24}$$

where $L_t \le \exp\{C2^{t/2}\}$. The finite prefactor $(k-1)/\sqrt{k-2}$ in the interpolation differs from $\sqrt{k}$ by a relative $O(k^{-1})$, which is included in this error.

Multiply (23) by λ_{t+1} and sum over root trajectories. The $s=t$ contribution has nonnegative weights:

$$\frac{\sqrt{k}}{2}\sum_{\lambda} d_t(\lambda)J^t_t(\lambda).$$

By (14), if $\eta_t = \sup_u |d_t(u)/b_t - 1|$, this lies between $(1-\eta_t)b_{t+1}$ and $(1+\eta_t)b_{t+1}$. Thus the previous relative error is not multiplied by a large condition number.

For $s<t$, use $J^t_s \le (2\pi)^{-1/2}$ and (22). After division by b_{t+1}, the total absolute contribution is at most $2^{T+C}\Lambda_T^2/\sqrt{k}$. Furthermore, the bootstrap gives $D_t \le 12\Lambda_T/k$ and $\sum_{s\le t} d_s(\lambda) \le 3b_t$. Proposition 7.1 bounds e_t by a controlled constant divided by $\sqrt{k}$. Summing (24) over the root trajectories and dividing by b_{t+1} is therefore also at most a controlled constant divided by $\sqrt{k}$. The initial-law term has this bound because $b_t \ge \theta$ and $r_{t+1}^{-1} \le \Lambda_T$. We have proved

$$\eta_{t+1} \le \eta_t + \frac{\exp\{C2^{T/2}\}}{\sqrt{k}}.$$

Summing over t preserves the controlled-constant form. The degree condition makes the resulting error less than $1/2$, closing the bootstrap and proving (21). □

8 Saturation and consensus

Proof of Theorem 1.1. First, the lower bound on a_T and a sufficiently large choice of A ensure that the initialization in (2) belongs to $(0,1)$. At time T, (21) gives, uniformly in the parent trajectory,

$$d_T(u) = \frac{8}{\sqrt{k}}\left(1 + O\left(\frac{e^{C2^{T/2}}}{\sqrt{k}}\right)\right).$$

At earlier times the normalized bias increments are at most a controlled constant divided by $\sqrt{k}$, by (22). Thus the child sum in the exact cavity recursion at horizon $T+1$ has Gaussian mean

$$\mu(\lambda) + 8\mathbf{e}_T + O_\infty\left(\frac{e^{C2^{T/2}}}{\sqrt{k}}\right)$$

and covariance C_T up to an error of the same size. Here $\mathbf{e}_T$ denotes the last coordinate unit vector. Applying the ordinary orthant CLT and Gaussian parameter comparison shows that the root trajectory through $T+1$, for every imposed parent, differs in total variation by at most a controlled constant divided by $\sqrt{k}$ from the cavity process with a field 8 at its last update only.

Its last spin is

$$\operatorname{sign}(H_T + 8), \qquad H_T = \eta(T) + \sum_{s<T} R(T,s)\tau(s).$$

Equation (8) gives $\mathbb{E}H_T^2 \le 4$. Hence

$$\mathbb{E}\operatorname{sign}(H_T + 8) \ge 1 - 2\mathbb{P}(H_T \le -8) \ge 1 - \frac{8}{64} = \frac{7}{8}.$$

For the degree constant sufficiently large, the approximation error implies

$$\mathbb{E}_{P^u_{k,\theta}}\sigma(T+1) \ge \frac{1}{2} \quad \text{for every imposed parent trajectory } u. \tag{25}$$

Now return to the full k-regular tree and fix a vertex v. In each of the k components obtained by removing v, run a lower process with the parent of that component fixed at -1 for all times. These processes are independent, and their spins at the neighbors of v are simultaneously at most the spins in the original process. By (25) their means at time $T+1$ are at least $1/2$. Hoeffding's inequality gives

$$\mathbb{P}_\theta\{\sigma_v(T+2) = -1\} \le \mathbb{P}\left\{\sum_{i=1}^k \sigma_{i,-}(T+1) \le 0\right\} \le e^{-k/8}.$$

This includes possible ties when k is even. Lemma 1.2 of [1] says that, for absolute $k_*, \delta_* > 0$, $\mathbb{E}_\theta\sigma_v(t) > 1 - \delta_*/k$ at some finite time implies $\theta_*(k) \le \theta$. Our estimate implies this criterion for all sufficiently large k. For each fixed k, the selected $T = T(k)$ is finite, so the use of that criterion entails no uniformity in the finite-graph limit used in its proof. Finally, the bound on a_T in (12) proves (3). □

Proof of Theorem 1.2. Let $L = \log k$ and $\ell = \log L$. Choose a fixed, sufficiently large constant b and set

$$T = \left\lfloor \frac{2}{\log 2}(\ell - b) \right\rfloor.$$

For sufficiently large k, this is nonnegative and

$$2^{T/2} \le Ce^{-b}L.$$

Choose b so that (1) holds. Theorem 1.1 then gives

$$\begin{aligned}\log\theta_*(k) &\le -\frac{T+1}{2}L + B2^{T/2}\\ &\le -\frac{L}{\log 2}\ell + O(L).\end{aligned}$$

When $\theta_*(k) = 0$, the same conclusion is read directly as the upper bound on $\theta_*(k)$ rather than as an assertion about its logarithm. This proves (4). □

9 Scope of the estimate

The proof establishes an explicit upper bound for the original dynamics, not for an independent-neighbor surrogate. It uses monotonicity only in valid comparisons with imposed parent trajectories. It retains the response term generated by backtracking. The Gaussian local limit argument conditions on one coordinate count, and consequently does not require a lower bound on the probability of every complete trajectory. The bias comparison uses the exact interpolation (15); no neglected higher-order product expansion is hidden in the critical $k^{-1/2}$ stage.

The estimate does not show that the threshold is positive for large k, does not provide a matching lower bound, and does not identify its true decay scale. A faster explicit upper bound would follow from stronger long-time lower bounds on $R(t+1,t)$ and on $\lambda_{\min}(C_T)$. For example, polynomial-exponential control $\lambda_{\min}(C_T), r_T \ge e^{-CT^q}$, together with the corresponding quantitative propagation above, would permit a power-of-$\log k$ time horizon instead of a $\log\log k$ horizon. Such stronger cavity estimates are not assumed or proved here.

References

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